No.	Status	Date
LNDE-RPT-20220906-033-DS-4-RA-101	Issued	2022-09-16

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101 located within the ATU-100. The scope of work is limited to the thickness survey of the rich amine service piping to support the 3-year RBI survey requirements.

Marcus Chen, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of data. All activities were performed in accordance with the requirements of the Hamilton Plant asset integrity program.

Technical Basis

Thickness measurements were performed under QSCP-NDT-001 Rev 2. The inspection methodology and equipment calibration were conducted in accordance with ASTM E797.

The data recorded herein supports the findings and conclusions detailed in LNDE-RPT-20220906-033-R0.

Line Specifications

The subject line, 4-RA-101, is constructed from A106 Gr.B with a nominal outside diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design limits for this line are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading below this value is subject to the site NCR process.

Measurement Data

Thickness readings were taken at designated CMLs. As noted in LNDE-RPT-20220906-033-R0, CML 4-RA-101-06 was recorded as inaccessible due to the absence of scaffolding and will be carried forward to the 2024 survey.

The measurements were performed using the standard carbon steel velocity setting appropriate for A106 Gr.B material.

Thickness readings, measured 2022-09-16

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2019-10-18	5.93	5.82	0.049	0.053	42.6	2025-09-16	
6-SG-101-02	7.11	3.56	2019-10-18	5.51	5.42	0.061	0.052	30.3	2025-09-16	
6-SG-101-03	7.11	3.56	2019-10-18	5.56	5.29	0.067	0.088	19.7	2025-09-16	
6-SG-101-04	7.11	3.56	2019-10-18	5.34	5.19	0.072	0.059	22.8	2025-09-16	
6-SG-101-05	7.11	3.56	2019-10-18	5.04	4.72	0.091	0.100	11.6	2025-09-16	
6-SG-101-06	7.11	3.56	2019-10-18	5.56	5.55	0.057	0.050	34.7	2025-09-16	
6-SG-101-07	7.11	3.56	2019-10-18	6.24	6.16	0.027	0.027	>50	2025-09-16	
6-SG-101-08	7.11	3.56	2019-10-18	5.77	5.57	0.066	0.065	30.6	2025-09-16	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2019-10-18	5.05	4.77	0.089	0.099	12.3	2025-09-16	
6-SG-102-02	7.11	3.56	2019-10-18	5.10	4.64	0.093	0.120	9.0	2025-09-16	
6-SG-102-03	7.11	3.56	2019-10-18	4.92	4.69	0.089	0.080	12.8	2025-09-16	
6-SG-102-04	7.11	3.56	2019-10-18	4.90	4.62	0.089	0.085	11.9	2025-09-16	
6-SG-102-05	7.11	3.56	2019-10-18	6.07	5.96	0.044	0.035	>50	2025-09-16	
6-SG-102-06	7.11	3.56	2019-10-18	6.03	5.79	0.054	0.041	41.6	2025-09-16	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2019-10-18	7.29	6.83	0.064	0.152	19.9	2025-09-16	
4-RA-101-02	8.56	3.81	2019-10-18	7.39	6.79	0.066	0.174	17.1	2025-09-16	
4-RA-101-03	8.56	3.81	2019-10-18	6.64	5.98	0.099	0.253	8.6	2025-09-16	
4-RA-101-04	8.56	3.81	2019-10-18	7.54	7.21	0.054	0.120	28.4	2025-09-16	
4-RA-101-05	8.56	3.81	2019-10-18	7.18	6.73	0.071	0.165	17.7	2025-09-16	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	—	—	—	—	—	inaccessible – no reading
4-RA-101-07	8.56	3.81	2019-10-18	7.68	7.50	0.041	0.097	38.0	2025-09-16	
4-RA-101-08	8.56	3.81	2019-10-18	6.45	5.55	0.118	0.267	6.5	2025-09-16	
4-RA-101-09	8.56	3.81	2019-10-18	7.68	7.37	0.045	0.115	30.9	2025-09-16	
4-RA-101-10	8.56	3.81	2019-10-18	7.55	7.19	0.056	0.126	26.8	2025-09-16	
4-RA-101-11	8.56	3.81	2019-10-18	6.66	6.07	0.098	0.232	9.7	2025-09-16	
4-RA-101-12	8.56	3.81	2019-10-18	7.03	6.41	0.213	0.213	12.2	2025-09-16	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2019-10-18	7.58	7.42	0.042	0.091	39.7	2025-09-16	
4-RA-102-02	8.56	3.81	2019-10-18	6.96	6.43	0.081	0.184	14.3	2025-09-16	
4-RA-102-03	8.56	3.81	2019-10-18	7.05	6.56	0.075	0.150	18.3	2025-09-16	
4-RA-102-04	8.56	3.81	2019-10-18	7.04	6.64	0.072	0.164	17.3	2025-09-16	
4-RA-102-05	8.56	3.81	2019-10-18	7.06	6.56	0.075	0.168	16.3	2025-09-16	
4-RA-102-06	8.56	3.81	2019-10-18	7.15	6.65	0.070	0.164	17.3	2025-09-16	
4-RA-102-07	8.56	3.81	2019-10-18	7.30	6.85	0.065	0.141	21.6	2025-09-16	
4-RA-102-08	8.56	3.81	2019-10-18	6.40	5.86	0.099	0.226	9.1	2025-09-16	
4-RA-102-09	8.56	3.81	2019-10-18	6.78	6.42	0.080	0.170	15.4	2025-09-16	
4-RA-102-10	8.56	3.81	2019-10-18	7.36	6.86	0.172	0.172	17.8	2025-09-16	
4-RA-102-11	8.56	3.81	2019-10-18	7.72	7.33	0.074	0.096	36.8	2025-09-16	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2019-10-18	9.83	9.55	0.055	0.112	44.4	2025-09-16	
6-RA-103-02	10.97	4.57	2019-10-18	8.61	7.97	0.118	0.261	13.0	2025-09-16	
6-RA-103-03	10.97	4.57	2019-10-18	9.49	8.95	0.077	0.168	26.0	2025-09-16	
6-RA-103-04	10.97	4.57	2019-10-18	8.22	7.30	0.139	0.305	9.0	2025-09-16	
6-RA-103-05	10.97	4.57	2019-10-18	8.33	7.43	0.133	0.311	9.2	2025-09-16	
6-RA-103-06	10.97	4.57	2019-10-18	9.79	9.45	0.059	0.132	37.0	2025-09-16	
6-RA-103-07	10.97	4.57	2019-10-18	9.59	9.15	0.069	0.159	28.8	2025-09-16	
6-RA-103-08	10.97	4.57	2019-10-18	9.73	9.29	0.063	0.158	29.9	2025-09-16	
6-RA-103-09	10.97	4.57	2019-10-18	9.78	9.43	0.060	0.137	35.6	2025-09-16	
6-RA-103-10	10.97	4.57	2019-10-18	8.37	7.68	0.126	0.275	11.3	2025-09-16	
6-RA-103-11	10.97	4.57	2019-10-18	8.91	—	—	—	—	—	inaccessible – no reading
6-RA-103-12	10.97	4.57	2019-10-18	9.52	9.11	0.141	0.141	32.3	2025-09-16	
6-RA-103-13	10.97	4.57	2019-10-18	9.55	9.13	0.114	0.155	29.5	2025-09-16	
6-RA-103-14	10.97	4.57	2019-10-18	8.26	7.34	0.316	0.316	8.8	2025-09-16	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2019-10-18	6.05	5.92	0.063	0.064	37.0	2025-09-16	
3-RV-101-02	7.62	3.56	2019-10-18	5.97	5.81	0.066	0.076	29.7	2025-09-16	
3-RV-101-03	7.62	3.56	2019-10-18	5.24	4.98	0.096	0.099	14.4	2025-09-16	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-04	7.62	3.56	2019-10-18	5.58	5.39	0.083	0.074	22.1	2025-09-16	
3-RV-101-05	7.62	3.56	2019-10-18	6.11	5.91	0.066	0.068	34.4	2025-09-16	
3-RV-101-06	7.62	3.56	2019-10-18	5.55	5.39	0.083	0.086	21.2	2025-09-16	
3-RV-101-07	7.62	3.56	2019-10-18	5.07	4.80	0.117	0.112	10.6	2025-09-16	
3-RV-101-08	7.62	3.56	2019-10-18	5.86	5.81	0.075	0.076	29.7	2025-09-16	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2019-10-18	4.16	3.91	0.061	0.067	16.8	2025-09-16	
2-RV-102-02	5.54	2.79	2019-10-18	3.86	3.64	0.074	0.056	11.6	2025-09-16	
2-RV-102-03	5.54	2.79	2019-10-18	4.37	4.34	0.043	0.046	34.1	2025-09-16	
2-RV-102-04	5.54	2.79	2019-10-18	4.59	4.47	0.042	0.039	40.1	2025-09-16	
2-RV-102-05	5.54	2.79	2019-10-18	4.78	4.59	0.037	0.058	31.2	2025-09-16	
2-RV-102-06	5.54	2.79	2019-10-18	4.27	4.05	0.064	0.033	19.6	2025-09-16	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2019-10-18	7.15	6.61	0.072	0.167	16.8	2025-09-16	
4-RA-104-02	8.56	3.81	2019-10-18	7.31	7.04	0.057	0.123	26.3	2025-09-16	
4-RA-104-03	8.56	3.81	2019-10-18	7.94	7.70	0.074	0.088	44.2	2025-09-16	
4-RA-104-04	8.56	3.81	2019-10-18	7.64	7.31	0.101	0.132	26.5	2025-09-16	
4-RA-104-05	8.56	3.81	2019-10-18	7.57	7.12	0.109	0.158	21.0	2025-09-16	
4-RA-104-06	8.56	3.81	2019-10-18	7.13	6.70	0.071	0.138	20.9	2025-09-16	
4-RA-104-07	8.56	3.81	2019-10-18	6.67	6.04	0.096	0.194	11.5	2025-09-16	
4-RA-104-08	8.56	3.81	2019-10-18	7.46	7.09	0.094	0.109	30.0	2025-09-16	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2022-09-16	_____